

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Thomsh. K (19241004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thmsh. K

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SP

Smart Home Automation System

Introduction:

A smart home automation system controls lighting, temperature & using sensors. It responds to real-time inputs (motion, temperature, time) and learn preferences

Types of Intelligent Agents

1. Simple Reflex Agent

- * Acts only on current sensor input
- * Uses condition-action rules

Eg:

- * If motion detected → Turn On lights

2. Model-Based Agent

- * Uses memory of previous states

Eg:

- * Keeps lights ON if someone is still in the room

3. Goal-Based Agent

- * Acts to achieve goals like comfort, security & energy saving

Eg:

- * Maintains room temperature at 24°C

4. Utility-Based Agent

- * Chooses the best action

is not intended appliances to save power

Agent	Function
Reflex	Responds to current sensor input
Model-Based	Uses memory to previous states
Goal-Based	Achieves specific goals
Utility-Based	Selects the best overall action

Conclusion:
A Hybrid Agent (Model-Based + Goal-Based + Utility-Based) is the most effective

- Example it:
- * Learns user performances
 - * Maintains home comfort
 - * Saves energy
 - * Improves security
 - * Makes optimal decisions in different situations

Conclusion:
A Hybrid intelligent agent provides the best balance between
... security.

... intelligent system controls traffic lights and
... density, pedestrian movement, and weather conditions to reduce car
... and time.

+ Intelligent Agents

... Reflex Agent

... only on current traffic conditions

... and

... structure

+ Heavy traffic detected $\rightarrow$ Green signal extended

+ No vehicle $\rightarrow$ Red signal

Full-Bel Agent

... previous traffic information

... and

... use traffic history to predict congestion during peak hours

... Goal Agent

... towards specific goals

... and

+ Reduce waiting time

+ ... traffic congestion

4. utility - Based Agent

selects the action that gives maximum overall benefit

Examples:

If an ambulance is detected, the system gives immediate green signals.

Comparison

Agent	Function
Simple - Reflex	Reacts to current traffic
Model Based	Uses previous traffic data
Goal Based	Achieves traffic management goals
utility - Based	selects the best overall action

Justification :

A Hybrid Agent (Model - Based + Based - goal + utility - Based) is the most suitable because it: Learns traffic patterns; Predicts congestion; Reduces travel time

Conclusion:

A Hybrid intelligent agent is the best solution for minimizing congestion and travel time while ensuring efficient traffic control.